# DSC 291 Learning Theory - PA4

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## Part A: Sparse Linear Predictors and Model Selection

### 1. VC Dimension Through Supports

**Statement.** For the class $\mathcal{H}_k$ of $k$-sparse homogeneous halfspaces in $\mathbb{R}^d$, the VC dimension satisfies:

$$\mathrm{VCdim}(\mathcal{H}_k) = O\big(k \log(\tfrac{ed}{k})\big)$$

**Proof.** Count possible supports, apply the VC growth function bound to each $k$-dimensional subclass, and union-bound the total growth function.

1. **Decomposition by Support:** Let $I \subseteq [d]$ with $|I| = k$ be a specific support. The subclass $\mathcal{H}_I$ (halfspaces restricted to coordinates in $I$) is isomorphic to homogeneous halfspaces in $\mathbb{R}^k$. The VC dimension of each subclass is $k$.
2. **Subclass Growth Function:** By Sauer's Lemma, the growth function for any subclass $\mathcal{H}_I$ is bounded by:

$$\Pi_{\mathcal{H}_I}(n) \leq \left(\frac{en}{k}\right)^k$$

3. **Union Bound on Growth Functions:** The class $\mathcal{H}_k$ is the union of $\binom{d}{k}$ such subclasses. The growth function of a union is bounded by the sum of the growth functions of its components:

$$\Pi_{\mathcal{H}_k}(n) \leq \sum_{I:|I|=k} \Pi_{\mathcal{H}_I}(n)$$

   Substituting the bound from Step 2, this leads to the inequality:

$$2^n \leq \binom{d}{k}\left(\frac{en}{k}\right)^k$$

4. **Solving for $n$:** To find the maximum $n$ that can be shattered, we take the logarithm of both sides and use the inequality $\log\binom{d}{k} \leq k\log(\frac{ed}{k})$:

$$n \leq \log_2\binom{d}{k} + k\log_2\left(\frac{en}{k}\right) \leq k\log_2\left(\frac{ed}{k}\right) + k\log_2\left(\frac{en}{k}\right)$$

5. **Conclusion:** For large $n$, the linear term $n$ on the left is constrained by the $k\log(d/k)$

term on the right. Solving this transcendental inequality yields $n = O(k \log(\frac{ed}{k}))$. Thus:

$$\mathrm{VCdim}(\mathcal{H}_k) = O\big(k \log(\tfrac{ed}{k})\big) \quad \blacksquare$$

## 2. Penalty-based SRM

**Statement.**

With a prior $p_k$ over sparsity levels $k = 1, \ldots, d$ (where $\sum p_k \leq 1$), define the SRM estimator $\hat{h}$ as:

$$\hat{h} \in \arg\min_{k, h \in \mathcal{H}_k} \left\{ L_S(h) + C\sqrt{\frac{k \log(\frac{ed}{k}) + \log(\frac{1}{p_k}) + \log(\frac{1}{\delta})}{n}} \right\}$$

With probability $\geq 1 - \delta$, the generalization error satisfies:

$$L_{\mathcal{D}}(\hat{h}) \leq \inf_{1 \leq k \leq d,\, h \in \mathcal{H}_k} \left\{ L_{\mathcal{D}}(h) + 2C\sqrt{\frac{k \log(\frac{ed}{k}) + \log(\frac{1}{p_k}) + \log(\frac{1}{\delta})}{n}} \right\}$$

**Proof.** Instantiate the class-level SRM theorem using the VC dimensions for sparse halfspaces.

1. **Class-Level SRM:** General SRM theory guarantees that for a collection of classes $\{\mathcal{H}_k\}$, an oracle inequality holds if the penalty accounts for the complexity of each $\mathcal{H}_k$ and the prior weight $p_k$ assigned to it.
2. **Substituting Complexity:** From Part 1, we know the VC dimension of $k$-sparse halfspaces in $d$ dimensions is $\mathrm{VCdim}(\mathcal{H}_k) \leq C_1 k \log(\frac{ed}{k})$.
3. **Decomposing the Penalty:** The term inside the square root represents a triple union bound:
   - **Support-selection cost:** $k \log(\frac{ed}{k})$ pays for searching over $\binom{d}{k}$ possible coordinate sets.
   - **Sparsity-level cost:** $\log(1/p_k)$ pays for the choice of $k$ itself.
   - **Confidence cost:** $\log(1/\delta)$ ensures the bound holds with high probability.
4. **Conclusion:** Combining these terms and absorbing universal constants into $C$ yields the stated bound. $\blacksquare$

**Statistical Cost Paid for Choosing Support vs. Sparsity Level**

The SRM framework reveals a hierarchy of "prices" that must be paid to ensure generalization when searching for sparse patterns:

- **The Support vs. The Level:** There is a fundamental distinction between the cost of finding the *right features* and the cost of choosing the *number of features*. The $k \log(d/k)$ term dominates when $d$ is large, as it represents the massive combinatorial search space of supports.

- **The Power of the Prior:** By choosing a non-uniform prior (e.g., $p_k \propto 1/k^2$), we can assign more "budget" to simpler models. This prevents a uniform penalty of $\log d$ from over-penalizing very sparse solutions, allowing the algorithm to remain sensitive to simple structures even in extremely high dimensions.
- **Occam's Razor by Design:** The SRM objective functions as a mathematical realization of Occam's Razor. It will only accept a more complex, denser model (higher $k$) if the improvement in empirical error $L_S(h)$ is large enough to compensate for the increased statistical "cost" of the more complex support.

**Takeaway.** SRM efficiently manages the trade-off between fitting the data and the double-complexity of selecting both the sparsity level and the specific support, preferring smaller supports unless the data provides overwhelming evidence for density.

## 3. Comparison with Dense Halfspaces

**Statement.** If $\exists w^*$ with $\|w^*\|_0 \le s$ and $L_{\mathcal{D}}(w^*) \le \eta$, then any $n$ satisfying

$$n \gtrsim \frac{s\log(\frac{ed}{s}) + \log(\frac{1}{p_s}) + \log(\frac{1}{\delta})}{\epsilon^2}$$

ensures $L_{\mathcal{D}}(\hat{h}) \le \eta + \epsilon$ w.p. $\ge 1-\delta$. Dense halfspaces need $n \gtrsim \frac{d+\log(1/\delta)}{\epsilon^2}$. Sparse bound is smaller when $s\log(\frac{ed}{s}) \ll d$.

**Proof.** Plug $h = w^*$ into Part 2 bound and solve for $n$; compare with VC($d$) rate for dense halfspaces.

1. Part 2 with $k = s$ yields $L_{\mathcal{D}}(\hat{h}) \le \eta + C\sqrt{\frac{s\log(\frac{ed}{s})+\log(1/p_s)+\log(1/\delta)}{n}}$.
2. Enforce the square-root term $\le \epsilon$ to obtain the stated $n$.
3. For dense halfspaces, $\mathrm{VCdim} = d$, giving rate $\sqrt{\frac{d+\log(1/\delta)}{n}}$.
4. Sparse advantage holds when $s\log(\frac{ed}{s})$ grows sublinearly vs. $d$. ■

**Comparison to the sample size obtained by learning over all homogeneous halfspaces in $\mathbb{R}^d$.**

- Dense halfspaces need $n \gtrsim \frac{d+\log(1/\delta)}{\epsilon^2}$.
- The sparse bound needs $n \gtrsim \frac{s\log(\frac{ed}{s})+\log(1/p_s)+\log(1/\delta)}{\epsilon^2}$.
- Sparse wins whenever $s\log(\frac{ed}{s}) \ll d$), e.g., $(s = O(\log d))$ or $(s = d^\alpha)$ with $(\alpha < 1)$.
- SRM can to adapt to unknown sparsity, provided the penalty properly encodes the search over supports

## 4. Validation as Model Selection

**Statement.** Using split $S_1, S_2$ with $|S_1| = |S_2| = n/2$, ERM $w_k$ on $S_1$ for each $k$, and $\hat{k}$ minimizing $L_{S_2}(w_k)$, we have w.p. $\geq 1 - \delta$,

$$L_{\mathcal{D}}(w_{\hat{k}}) \leq \inf_{1 \leq k \leq d,\ \|w\|_0 \leq k} \left\{ L_{\mathcal{D}}(w) + C\sqrt{\frac{k \log(\frac{ed}{k}) + \log(\frac{d}{\delta})}{n}} \right\}.$$

**Proof.** Condition on $S_1$; finite-class validation on $S_2$ plus VC control of ERM on $S_1$.

1. Condition on $S_1$. The $d$ candidates $\{w_k\}$ form a finite class for $S_2$.
2. [Ref: Weighted finite-class bound] with uniform weights $1/d$ on $S_2$: w.p. $\geq 1 - \delta$, for all $k$, $L_{\mathcal{D}}(w_k) \leq L_{S_2}(w_k) + \sqrt{\frac{\log d + \log(1/\delta)}{n}}$.
3. Optimality of $\hat{k}$ on $S_2$ gives $L_{S_2}(w_{\hat{k}}) \leq L_{S_2}(w_k)$ for all $k$.
4. VC generalization on $S_1$: w.p. $\geq 1 - \delta$, $L_{\mathcal{D}}(w_k) \leq L_{S_1}(w_k) + C'\sqrt{\frac{k \log(\frac{ed}{k}) + \log(1/\delta)}{n}}$, and ERM implies $L_{S_1}(w_k) \leq L_{S_1}(w)$ for any $\|w\|_0 \leq k$.
5. Combine (2)-(4) with $|S_1| = |S_2| = n/2$ and union-bound $\delta$'s; absorb constants to reach the displayed bound. Cost $k \log(\frac{ed}{k})$ fits $w_k$; $\log d$ selects $k$. ■

**Comparison to penalty-based SRM.**

- **Data Usage & Fitting: SRM** uses the full sample $S$ of size $n$ to fit every hypothesis $h \in \mathcal{H}_k$. **Validation** splits the data, using only $S_1$ (size $n/2$) to fit the candidates $\{w_k\}$.
- **Complexity Selection: SRM** chooses $k$ by minimizing the penalized empirical risk (training error + penalty) on the full sample. **Validation** chooses $k$ by minimizing the holdout risk specifically on the unseen set $S_2$.
- **Statistical Price: Validation** pays a "sample size tax"; because it uses $n/2$ for both fitting and selection, the variance in the bounds increases by a factor of $\sqrt{2}$. It also incurs a $\log d$ cost to account for the finite choice among $d$ models.
- **SRM Trade-off: SRM** maintains higher efficiency by using all $n$ points for both tasks, but it pays a "complexity tax" via the $\log(1/p_k)$ term to avoid overfitting the combined classes.
- **Practical Advantage: Validation** is highly robust because it avoids the need to manually specify priors ($p_k$) or tune penalty constants. However, this simplicity comes at the cost of statistical efficiency, which is most noticeable when $n$ is small.

## 5. Computation of Sparse Feasibility

**Statement.** Brute-force realizable learning enumerates all supports $I \subseteq [d]$, $|I| = k$, solving halfspace feasibility on each. Runtime $= \binom{d}{k} T_{\text{feas}}(k, n)$; polynomial in $d$ only when $k$ is $O(1)$ or $O(\log d)$.

**Proof.** Count supports; multiply by per-support LP/feasibility cost.

1. Enumerate $\binom{d}{k}$ supports; for each, solve feasibility in $\mathbb{R}^k$ with $n$ constraints via LP in

$\mathrm{poly}(k, n)$.

2. Total time $\binom{d}{k}\,\mathrm{poly}(k,n) \approx (\frac{ed}{k})^k\,\mathrm{poly}(k,n)$.
3. Polynomial in $d$ only when $k$ is constant or logarithmic.
4. Highlights gap: sample complexity scales with $k\log(\frac{ed}{k})$, but computational cost can be exponential in $k$. ■

**Brute-force algorithm.** For each subset $I \subseteq [d]$ with $|I| = k$ (there are $\binom{d}{k}$ of them): restrict every $x_i$ to coordinates in $I$ and solve the linear feasibility problem:

$$\text{Find } w_I \in \mathbb{R}^k \text{ s.t. } y_i\langle w_I, x_i|_I\rangle > 0, \quad \forall i.$$

Finally, output any feasible $w_I$ after lifting back to $\mathbb{R}^d$ by padding zeros outside $I$.

**Runtime estimate.** Enumerating supports costs $\binom{d}{k}$. Feasibility for a fixed $I$ can be solved in time polynomial in $n$ and $k$ (e.g., via LP in $O(\mathrm{poly}(n,k))$). Total runtime

$$T(d,k,n) = \binom{d}{k} \cdot \mathrm{poly}(n,k).$$

This is polynomial in $d$ iff $k = O(1)$ or $k = O(\log d)$; once $k = \Theta(d)$ the binomial term is exponential in $d$.

**Distinction between sample complexity and computational complexity.**

Sample complexity depends on $k\log(\frac{ed}{k})$ (mild for moderate $k$), but brute-force enumeration is $\binom{d}{k} \cdot \mathrm{poly}(k,n)$, exponential in $k$; thus statistical feasibility does not imply computational tractability.

# Part B: PAC-Bayes for Thresholds

## 1. VC Dimension and Point-Posterior PAC-Bayes

**Statement.** For thresholds $\mathcal{H}_N$ on $\{1,\dots,N\}$, $\mathrm{VCdim} = 1$. With uniform prior $P$ and point posterior $Q_S = \delta_{\hat{h}_t}$ from a consistent ERM,

$$KL(Q_S \parallel P) = \log(N+1), \quad L_{\mathcal{D}}(Q_S) \le L_S(Q_S) + \sqrt{\frac{\log(N+1)+\log(\frac{2n}{\delta})}{2(n-1)}}.$$

**Proof.** Show VC=1; compute KL for point mass under uniform prior; plug into [Ref: PAC-Bayes].

1. Two labeled points with opposite labels cannot both be realized by a single threshold unless ordered; only single point shatterable, so VC=1.
2. Uniform prior: $P(h_t) = 1/(N+1)$; point posterior gives $KL = \log(N+1)$.
3. Realizable $S$ gives $L_S(Q_S) = 0$; substitute into [Ref: PAC-Bayes].

4. The bound scales with $\log(N+1)$ because KL accounts for hypothesis count, unlike VC realizable bounds that depend solely on VC=1. ■

**Why the PAC-Bayes certificate scales with $log(N+1)$ even though a VC-style realizable guarantee for thresholds should not scale with $log(N)$.**

PAC-Bayes must pay for the description length of the selected hypothesis under the fixed prior. VC realizable bounds do not, because they rely on growth functions rather than information-theoretic weights. Hence the mismatch: even though thresholds are extremely simple, a naive uniform prior distributes mass across $N+1$ hypotheses and incurs $\log(N+1)$ complexity.

## 2. Version-Space Posterior

**Statement.**

Let $a(S) = \max\{x_i : y_i = 0\}$ (0 if none) and $b(S) = \min\{x_i : y_i = 1\}$ ($N+1$ if none).

Then $V(S) = \{t : a(S) < t \leq b(S)\}$ with $|V(S)| = b(S) - a(S)$.

For uniform prior $P$, uniform posterior $Q_V$ over $V(S)$ satisfies $KL(Q_V \parallel P) = \log(N+1) - \log|V(S)|$ and $L_S(Q_V) = 0$.

**Proof.** Characterize consistency interval; compute KL of uniform-on-subset; note zero empirical loss.

1. Consistency requires threshold above largest negative and at/below smallest positive, giving the interval form.
2. $Q_V(h_t) = 1/|V(S)|$, $P(h_t) = 1/(N+1)$; KL simplifies to $\log(N+1) - \log|V(S)|$.
3. All $h_t$ in $V(S)$ classify $S$ correctly, so $L_S(Q_V) = 0$.
4. Larger $|V(S)|$ lowers KL versus a point posterior, tightening the PAC-Bayes certificate. ■

**Why $QV$ can give a better PAC-Bayes bound than a point posterior when $|V(S)|$ is large.**

When $|V(S)|$ is large, spreading the posterior uniformly over $V(S)$ reduces the KL term by $\log|V(S)|$ while keeping $L_S(Q_V) = 0$, tightening the PAC-Bayes bound relative to a point posterior.

## 3. Spreading Helps KL, May Hurt True Risk

**Statement.** For realizable $\mathcal{D}$ with true threshold $h_\tau$ (uniform $X$), any nonempty $W \subseteq \{1, \ldots, N+1\}$ and uniform posterior $Q_W$ satisfy

$$L_{\mathcal{D}}(Q_W) = \frac{1}{N|W|} \sum_{t \in W} |t - \tau|.$$

There exist realizable $S$ with large $|V(S)|$ such that $KL(Q_{V(S)} \parallel P) < KL(\delta_{h_\tau} \parallel P)$ yet $L_{\mathcal{D}}(Q_{V(S)}) > L_{\mathcal{D}}(\delta_{h_\tau})$.

**Proof.** Compute disagreement mass; exhibit wide interval around $\tau$.

1. Disagreement of $h_t$ vs $h_\tau$ occurs on indices between $t$ and $\tau$, fraction $|t-\tau|/N$; averaging gives displayed risk.
2. Example: $N = 20$, $\tau = 10$, and realizable $S$ leaving $V(S) = \{6, \ldots, 14\}$ (size 9). Then $KL(Q_{V(S)} \parallel P) = \log 21 - \log 9 < \log 21 = KL(\delta_{h_{10}} \parallel P)$.
3. Risk: $\frac{1}{20 \cdot 9} \sum_{t=6}^{14} |t-10| = \frac{53}{180} > 0$, while $L_{\mathcal{D}}(\delta_{h_{10}}) = 0$.
4. Lower KL can coexist with higher true risk; PAC-Bayes bounds certify via KL, not monotonic in true risk across posteriors. ■

**Why there is no contradiction with PAC-Bayes.**

A posterior with lower $KL$ can still have higher true risk if it spreads mass over suboptimal thresholds; PAC-Bayes bounds are driven by $KL$, not by monotonicity in true risk, so the example does not contradict PAC-Bayes. PAC-Bayes upper-bounds $L_{\mathcal{D}}(Q_W)$ by empirical loss plus a complexity term. Here $L_S(Q_{V(S)}) = 0$, and the complexity term remains nontrivial (though smaller than the point posterior). The theorem never promised that minimizing KL would minimize true risk—only that both terms together bound the risk.

## 4. Every Fixed Prior Can Be Attacked

**Statement.** For any prior $P$ on $\mathcal{H}_N$ with $N \geq 3$, some $\tau \in \{2, \ldots, N\}$ has $P(h_\tau) \leq 1/(N-1)$. For distribution $\mathcal{D}_\tau$ with mass $1/2$ on $(\tau-1, 0)$ and $1/2$ on $(\tau, 1)$, with probability $\geq 1 - 2^{1-n}$ over $S \sim \mathcal{D}_\tau^n$, both support points appear, so $V(S) = \{\tau\}$ and any zero-empirical-error posterior equals $\delta_{h_\tau}$, yielding $KL(Q \parallel P) \geq \log(N-1)$.

**Proof.** Pigeonhole on prior mass; bound missing-support probability; uniqueness of consistent threshold fixes posterior.

1. Internal thresholds sum to $\leq 1$; pigeonhole gives some $\tau$ with $P(h_\tau) \leq 1/(N-1)$.
2. Each support point missing with prob. $(1/2)^n$; union bound gives both seen with prob. $\geq 1 - 2^{1-n}$.
3. When both seen, consistency forces $V(S) = \{\tau\}$; any $Q$ with $L_S(Q) = 0$ is $\delta_{h_\tau}$.
4. Then $KL(Q \parallel P) = \log \frac{1}{P(h_\tau)} \geq \log(N-1)$.
5. This limits zero-empirical-error PAC-Bayes certificates with fixed prior, not sample complexity of thresholds (VC=1). ■

**Why it is a limitation of zero-empirical-error PAC-Bayes certificates with a fixed prior, rather than a lower bound on the sample complexity of learning thresholds?**

The lower bound shows that with a fixed prior and zero empirical error, $KL$ can be forced to $\Omega(\log N)$ even though thresholds have $VC = 1$. It is a limitation on zero-error PAC-Bayes

certificates with a fixed prior, not a lower bound on the sample complexity of learning thresholds (which remains constant-VC learnable). This is not a lower bound on sample complexity for learning thresholds; efficient VC algorithms achieve perfect generalization with $O(\frac{\log(1/\delta)}{\epsilon})$ samples independent of $N$. Rather, it shows a limitation of PAC-Bayes certificates that insist on zero empirical error with a fixed prior: an adversary can force large KL, hence loose bounds. Allowing nonzero empirical loss or data-dependent priors can evade this limitation.

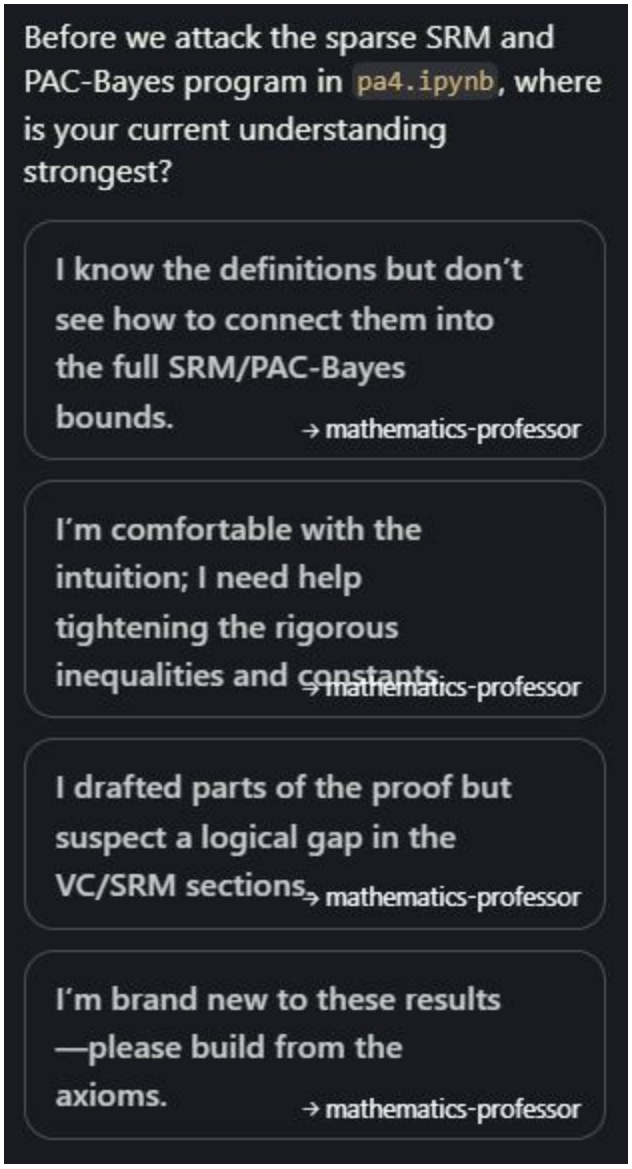
# Part C: AI Usage Report

1. Codex 5.1 was used to generate the inital proof sketches. that is, until Codex 5.1 was shown to offer the same price per tokens. So of course, all proofs were rewrtten from scratch with Codex 5.1 Max; only then were Codex 5.1's proofs appended.
   A. AI assisted workflow initially remained the same this week - create `AGENTS.md` and ask Codex 5.1 to build off what was written in `PLAN.md` under the role of "Professor". Due to the complexity of PAC-Bayes however, other Roo Code settings were explored.

   B. The key player for constructing proofs lies in Skills. Per the documentation:

      > Skills encapsulate contextual instructions and workflows, which the agent can load on-demand. They can include general instructions, specific workflows and even scripts to be run as needed. Here, a math proof skill was created using Gemini 3, which was built off the Role already assigned to Roo Code. The skill, stored under `~\.roo\skills\mathematical-proof-writer\SKILLS.md` served as a way to stardardize proof structures, which was important because Codex 5.1 was hallucinating on PAC-Bayes without guidance.

   C. To start, model reasoning was set to Low, previously it was not activated at all. This prompted Codex 5.1 to offer a variety of proof constructions, ranging from beginner to advanced. Selecting beginner prompted the following:




Builing from axioms only provided a surface level intuition of the theorems but did not explain the proof in ddetail. Moreover, it failed to provide a structured analysis of the algorithms, so Codex 5.1 Max rewrote everything.

D. Codex 5.1 Max maintained a consistent structure, making every proof follow the same format - statement, proof, reasoning. Its logic was easier to follow because it made use of the Skill written in (2). Due to the reasoning level being set to low, it also offered the following

Before we write the proofs, how rigorous and detailed do you want the derivations in pa4.ipynb?

- Full formal proofs with all inequalities derived explicitly
- Concise but rigorous sketches emphasizing key lemmas
- High-level outlines highlighting main steps only

**Thinking** 0s

Planning follow-up strategy

I'm deciding to use the ask_followup tool for baseline assessment and follow-ups, creating a todo list for task management due to complexity, and considering inspecting the pa4 notebook later.

**Updated the to-do list**

- → Review instructions in AGENTS.md and plan.md for theorem references
- [ ] Inspect pa4.ipynb relevant cells for Parts A & B
- [ ] Draft proofs for Part A (items 1-5) in notebook
- [ ] Draft proofs for Part B (items 1-4) in notebook
- [ ] Append notable blockers/nontrivial proofs to hw4/prompts/plan.md
- [ ] Final verification of notebook formatting and references

Interestingly, it made a "ask_followup" tool which was useful in clarifying how the proofs were to be drafted. It should be mentioned at this point that `PLAN.md` now contained only theorems and not blockers, as exercises are unrelated to each other. By placing the correct theorems to reference in `PLANS.md`, Codex 5.1 Max was able to finally generate proofs that ere more in line with mathematical reasoning instead of a high level intuition.

E. Still, the inution provided by Codex 5.1 was useful, and served as a better analysis that Codex 5.1 Max which only generated proofs. The analysis statements were clarified with Gemini 3 and underwent a final manual review to ensure that no logical jumps were taken and theorems were only taken from `PLANS.md` to build cohesive solutions. Statements from Codex 5.1 were then appended to the proofs from Codex 5.1 Max.

2. Most, if not all, proofs generated by Codex 5.1 Max were accepted due to its adherence to the Proof Skill in this workflow. There were no gaps in the output, due to a combination of the model reasoning now being set to Low (when previously it was set to None) and a reference sheet in `PLAN.md` that served as a better use of tokens than technical issues already solved in previous deliverables.
3. Most proofs were rejected from Codex 5.1 due to hallucinations of mistaken lemmas and incorrect notation that did not reference the new theorems in `PLAN.md`. While it only provided a high level explanation, it failed to make connections to previous work which further contributed to proofs without assumptions. Yet, it's intuitive analysis shined and thus was appended to the work done by Codex 5.1 Max, the latter failing to provide any analysis of the algorithms at all.
4. In addition to using Gemini 3 to verify the work done by Codex 5.1/Max, we also manually verified the proof structure referencing material from lecture. The "Researcher" role was not used in this workflow since it would likely return the same confirmation as the previous three AI agents.